(Robert Kohn)
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Portfolio Problems for Test 1

Exercise 1.3.3. *(a) Let A be nonempty and bounded below, and define*

$$B = \{b \in \mathbb{R} : b \text{ is a lower bound for } A\}.$$

Show that $\sup B = \inf A$.

Proof. Let $B = \{b : b \text{ is a lower bound of } A\}$. Since A is nonempty and bounded below, B is nonempty, and any element of A is an upper bound for B.

By the Axiom of Completeness, $s = \sup B$.

We show s is the greatest lower bound of A.

First, s is a lower bound for A. Suppose, for contradiction, that s is not a lower bound of A. Then there exists $a \in A$ with $a < s$. Set $\varepsilon = s - a > 0$.

By the approximation property of the supremum (Lemma 1.3.8), there exists $b \in B$ such that $s - \varepsilon < b \leq s$. But $s - \varepsilon = s - (s - a) = a$, so we have $a < b \leq s$.

Since $b \in B$, b is a lower bound of A. Yet $a \in A$ satisfies $a < b$, contradicting the fact that b is a lower bound of A. Hence s must be a lower bound of A.

Second, s is the greatest lower bound. Let ℓ be any lower bound of A. Then $\ell \in B$, and because $s = \sup B$, we have $\ell \leq s$. Thus s is greater than or equal to every lower bound of A, so $s = \inf A$.

$\therefore \sup B = \inf A$.

Exercise 1.4.7. *Finish the proof of the theorem by showing that the assumption $\alpha^2 > 2$ leads to a contradiction of the fact that $\alpha = \sup T$.*

Proof. Let $T = \{t \in \mathbb{R} : t^2 < 2\}$ and let $\alpha > 0$ satisfy $\alpha^2 > 2$. Observe that for any positive integer n,

$$\left(\alpha - \frac{1}{n}\right)^2 = \alpha^2 - \frac{2\alpha}{n} + \frac{1}{n^2} > \alpha^2 - \frac{2\alpha}{n}.$$

Choose $n \in \mathbb{N}$ such that:

$$\frac{1}{n} < \frac{\alpha^2 - 2}{2\alpha}.$$

Then

$$\alpha^2 - \frac{2\alpha}{n} > \alpha^2 - (\alpha^2 - 2) = 2,$$

and consequently

$$\left(\alpha - \frac{1}{n}\right)^2 > 2.$$

We now show that $\alpha - \frac{1}{n}$ is an upper bound for T. Take any $b \in T$, so $b^2 < 2$. Because $(\alpha - \frac{1}{n})^2 > 2 > b^2$ and both $\alpha - \frac{1}{n} > 0$ and $b^2 < 2 < (\alpha - \frac{1}{n})^2$, the inequality $x^2 > y^2$ for positive x, y implies $x > y$. Thus if $b \geq 0$, we have $\alpha - \frac{1}{n} > b$. If $b < 0$, then trivially $b < 0 < \alpha - \frac{1}{n}$. Hence in all cases $b \leq \alpha - \frac{1}{n}$.

Since $\alpha - \frac{1}{n}$ is an upper bound for T, α cannot be the least upper bound of T.

$\therefore$ If $\alpha > 0$ and $\alpha^2 > 2$, then $\alpha \neq \sup T$.

Exercise 2.5.7. *Extend the result of Example 2.5.3 by proving that the sequence* (b^n) *converges to* 0 *if and only if* $-1 < b < 1$.

Proof. We prove the equivalence in two directions.

($\Rightarrow$) Assume $\lim_{n\to\infty} b^n = 0$.

If $|b| \geq 1$, then $|b^n| = |b|^n \geq 1$ for all n. If a limit $L = \lim_{n\to\infty} b^n$ existed, taking absolute values gives $|L| = \lim_{n\to\infty} |b^n| \geq 1$. Thus, $L \neq 0$, a contradiction.

Hence $|b| < 1$.

($\Leftarrow$) Assume $-1 < b < 1$.

Case 1: $0 \leq b < 1$**.** Then (b^n) is decreasing and bounded below by 0, so it converges to some $\ell \geq 0$. Consider the subsequence $b^{2n} = (b^n)^2$. Taking limits in $b^{2n} = (b^n)^2$ yields $\ell = \ell^2$, hence $\ell = 0$ or $\ell = 1$. Since b^n is decreasing and $b < 1$, we cannot have $\ell = 1$. Therefore $b^n \to 0$.

Case 2: $-1 < b < 0$**.** Then $|b| \in (0, 1)$. By Case 1, $|b|^n \to 0$. Because $-|b|^n \leq b^n \leq |b|^n$ for all n, the Squeeze Theorem yields $b^n \to 0$.

Case 3: $b = 0$ is trivial.

$\therefore \lim_{n\to\infty} b^n = 0$ if and only if $-1 < b < 1$.